

Force Vectors & Resultants (review)

$F_R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$ – parallelogram law, resultant magnitude (θ between F_1, F_2); $\frac{F_1}{\sin\theta_1} = \frac{F_2}{\sin\theta_2} = \frac{F_R}{\sin\theta_R}$ – law of sines gives its direction

$F_x = F\cos\theta$, $F_y = F\sin\theta$ – rectangular components (2D), θ from the x -axis

$F = \sqrt{F_x^2 + F_y^2}$, $\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right)$ – magnitude/direction from components

$\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}$, $F = \sqrt{F_x^2 + F_y^2 + F_z^2}$ – Cartesian vector notation (3D)

$\mathbf{u} = \frac{\mathbf{F}}{F} = \cos\alpha\mathbf{i} + \cos\beta\mathbf{j} + \cos\gamma\mathbf{k}$ – unit vector & direction cosines (α, β, γ from $+x, +y, +z$)

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ – direction cosine identity

$\mathbf{r} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$ – position vector, tail to head

$\mathbf{u} = \frac{\mathbf{r}}{r}$, $\mathbf{F} = F\mathbf{u}$ – force along a line

$\mathbf{F}_R = \sum \mathbf{F} = (\sum F_x)\mathbf{i} + (\sum F_y)\mathbf{j} + (\sum F_z)\mathbf{k}$ – resultant of a concurrent force system: sum components independently

$\mathbf{A} \cdot \mathbf{B} = A_xB_x + A_yB_y + A_zB_z = AB\cos\theta$ – dot product, θ between $\mathbf{A}, \mathbf{B}$; $\theta = \cos^{-1}\left(\frac{\mathbf{A} \cdot \mathbf{B}}{AB}\right)$; projection of $\mathbf{A}$ onto a line with unit vector $\mathbf{u}$ is $\mathbf{A} \cdot \mathbf{u}$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

– cross product; $|\mathbf{A} \times \mathbf{B}| = AB\sin\theta$, direction by right-hand rule. Used for $\mathbf{M} = \mathbf{r} \times \mathbf{F}$

Moments, Couples & Force-Couple Systems

$M_O = Fd$ – scalar moment, d = perpendicular distance from O to the line of action of F ; sign by right-hand rule (CCW + common)

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

– vector moment; $\mathbf{r}$ = position vector from O to any point on the line of action of $\mathbf{F}$ (preferred for 3D)

$\mathbf{M}_O = \sum(\mathbf{r} \times \mathbf{F}) = \sum(xF_y - yF_x)$ – Varignon's theorem: moment of a force = sum of moments of its components

$$M_a = \mathbf{u}_a \cdot (\mathbf{r} \times \mathbf{F}) = \begin{vmatrix} u_x & u_y & u_z \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

– moment about a specified axis (triple scalar product); $\mathbf{u}_a$ along the axis, $\mathbf{r}$ from any point on the axis to any point on the line of action

$M = Fd$, $\mathbf{M} = \mathbf{r} \times \mathbf{F}$ – moment of a couple; couple = two equal, opposite, non-collinear forces; its moment is the same about every point (free vector)

$\mathbf{F}_R = \sum \mathbf{F}$, $\mathbf{M}_{R_O} = \sum \mathbf{M}_O + \sum(\mathbf{r} \times \mathbf{F})$ – equivalent force-couple system at a chosen point O

$d = \frac{M_{R_O}}{F_R}$ – reduces to a single resultant force (no couple) only when $\mathbf{F}_R \perp \mathbf{M}_{R_O}$; line of action shifts by d from O

Equilibrium of Rigid Bodies & Dry Friction

$\sum F_x = 0$, $\sum F_y = 0$, $\sum M_O = 0$ – equilibrium, 2D (3 independent scalar eqns; alternate sets like $\sum M_A = \sum M_B = \sum M_C = 0$,

A, B, C non-collinear, also valid)

$\sum F_x = 0$, $\sum F_y = 0$, $\sum F_z = 0$, $\sum M_x = 0$, $\sum M_y = 0$, $\sum M_z = 0$ – equilibrium, 3D (6 independent scalar eqns; equivalently $\sum \mathbf{F} = 0$, $\sum \mathbf{M}_O = 0$)

Two-force member: loaded at only two points, no applied couples $\Rightarrow$ forces at both points equal, opposite, and collinear with the line joining them

Three-force member: equilibrium under exactly three forces requires them to be concurrent or all mutually parallel

2D support reactions: roller/rocker: 1 unknown, $\perp$ to the surface – pin/hinge: 2 unknowns (R_x, R_y), no moment – fixed support: 3 unknowns (R_x, R_y, M), no translation or rotation – cable/two-force link: 1 unknown, tension along the member – smooth surface contact: 1 unknown, normal force

3D support reactions: ball-and-socket: 3 unknowns (F_x, F_y, F_z), no moment – single journal bearing: 4 unknowns (2 $\perp$ forces + 2 $\perp$ moments) – single thrust bearing: 5 unknowns (2 $\perp$ forces + axial force + 2 $\perp$ moments) – fixed support: 6 unknowns ($F_x, F_y, F_z, M_x, M_y, M_z$) – smooth surface/roller: 1 unknown, normal force

$F \leq \mu_s N$ (no slipping), $F = \mu_k N$ (slipping) – dry (Coulomb) friction; μ_s, μ_k = static/kinetic coefficients ($\mu_k < \mu_s$ typically), N = normal force

$\phi_s = \tan^{-1}(\mu_s)$ – angle of friction; at impending slip the resultant contact force makes angle ϕ_s with the normal. Always check both slipping and tipping

Distributed Forces – Centroids & Loads

$$\bar{x} = \frac{\int x dA}{\int dA}, \quad \bar{y} = \frac{\int y dA}{\int dA}$$

– centroid of an area: average position of the dA elements

$$\bar{x} = \frac{\int x dL}{\int dL}, \quad \bar{y} = \frac{\int y dL}{\int dL} \text{ – centroid of a line (wire/rod/curve)}$$

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{\sum A_i}, \quad \bar{y} = \frac{\sum \bar{y}_i A_i}{\sum A_i} \text{ – composite bodies: locate each part's centroid, take the weighted average; subtract } A_i \text{ for holes}$$

$A = \theta \bar{r} L$, $V = \theta \bar{r} A$ – Pappus-Guldinus: surface area/volume of revolution; θ = angle of revolution (2π full turn), $\bar{r}$ = centroid distance from the axis

$F_R = \int w(x) dx$, $\bar{x} = \frac{\int x w(x) dx}{\int w(x) dx}$ – resultant of a distributed load on a beam: F_R = area under the load diagram, acts through its centroid

Centroids – common shapes: $\bar{x} = b/2$, $\bar{y} = h/2$ – rectangle ($b \times h$), geometric center – $\bar{y} = h/3$ – triangle centroid height, from the base – $\bar{x} = \bar{y} = 0$ – circle, at center – $\bar{y} = 4r/3\pi$ – semicircular area, from the diameter – $\bar{x} = \bar{y} = 4r/3\pi$ – quarter-circular area, from each straight edge

Centroids – lines & arcs: midpoint – straight line segment – $\bar{x} = \bar{y} = 2r/\pi$ – quarter-circular arc, from the center – $\bar{y} = 2r/\pi$ – semicircular arc, from the diameter

Trusses, Frames & Machines

$\sum F_x = 0$, $\sum F_y = 0$ (at each joint) – method of joints: isolate each pin as a particle; start where at most 2 unknowns act, work outward

$\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$ (on the cut section) – method of sections: cut up to 3 unknown members, keep one side, solve directly for a specific member force

Zero-force members: two non-collinear members at a joint with no load/support there $\Rightarrow$ both are zero-force – three members at a joint

(two collinear), no external load there $\Rightarrow$ the third (non-collinear) member is zero-force

$m + r = 2j$ – static determinacy of a plane truss; m = members, r = reaction components, j = joints. =: determinate, <: unstable, >: indeterminate

Frames/machines (multi-force members): dismember at the pins; draw a separate FBD for each member loaded at 3+ points; apply $\sum F_x = \sum F_y = \sum M = 0$ to each, using Newton's 3rd law at shared pins

Internal Forces in Members

$\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$ – internal loadings by the method of sections: cut at the section of interest, apply equilibrium to one segment for N, V, M

Sign convention: N positive in tension; V positive when it tends to rotate the segment clockwise; M positive when it bends the member concave up (sagging) – fix this before drawing N, V, M diagrams

Procedure: (1) whole-body FBD for reactions (2) cut at the section (3) FBD of one segment showing N, V, M with the sign convention (4) apply the three equilibrium equations

Same method-of-sections approach applies to each multi-force member of a frame/machine – cut the individual member, not just the whole structure, to expose its own N, V, M

$\sigma = \frac{N}{A}$ – internal axial force N produces an average normal stress over area A (bridge to Mechanics of Materials)

Quick Reference – at a glance

Equilibrium: 2D: $\sum F_x = \sum F_y = \sum M_O = 0$ (3 eqns) – 3D: $\sum F_x = \sum F_y = \sum F_z = \sum M_x = \sum M_y = \sum M_z = 0$ (6 eqns)

Support unknowns: roller/rocker 1 – pin/hinge 2 – fixed (2D) 3 – cable/link 1 – ball-and-socket 3 – journal bearing 4 – thrust bearing 5 – fixed (3D) 6

Centroids, from the named edge/axis: rectangle $b/2, h/2$ – triangle $h/3$ – circle 0 – semicircular area $4r/3\pi$ – quarter-circular area $4r/3\pi$ (both axes) – semicircular arc $2r/\pi$ – quarter-circular arc $2r/\pi$ (both axes)

Trusses & friction: $m + r = 2j$ determinate $F \leq \mu_s N$ (static)
 $F = \mu_k N$ (kinetic) $\phi_s = \tan^{-1} \mu_s$